CIE AS PHYSICS (9702) • QUESTION PAPER

6.1 Stress & Strain

Syllabus Section: 6. Deformation of Solids **Paper Type:** Multiple Choice (Paper 1) **Total:** 25 Questions (25 Marks)

Question 1 Easy [1 mark]

A student wanted to measure the Young modulus of a wire, to do this they needed to take a number of measurements.

In which row could the measurement **not** be made **directly** with the listed apparatus?

	Measurement	Apparatus
A	extension of wire	vernier scale
B	area of cross-section of wire	micrometer screw gauge
C	original length of wire	metre rule
D	mass of load applied to a wire	electronic balance

A

B

C

D

Question 2 Easy [1 mark]

What is the unit of the Young modulus?

A $N\ m^{-2}$

B N m

C $N\ m^{-1}$

D $N\ m^{2}$

Question 3 Easy [1 mark]

A material is extended elastically within the limit of proportionality. What is equal to the Young modulus of the material?

A gradient of the stress-strain graph

B area under the force-extension graph

C area under the stress-strain graph

D gradient of the force-extension graph

Question 4 Easy [1 mark]

An experiment was carried out to determine the Young modulus of steel using a steel wire.

The uncertainties in the measurements are given below.

Measurement	Uncertainty
load on wire	±2%
length of wire	±0.2%
diameter of wire	±1.5%
extension	±1%

What is the percentage uncertainty in the Young modulus?

A 1.3%

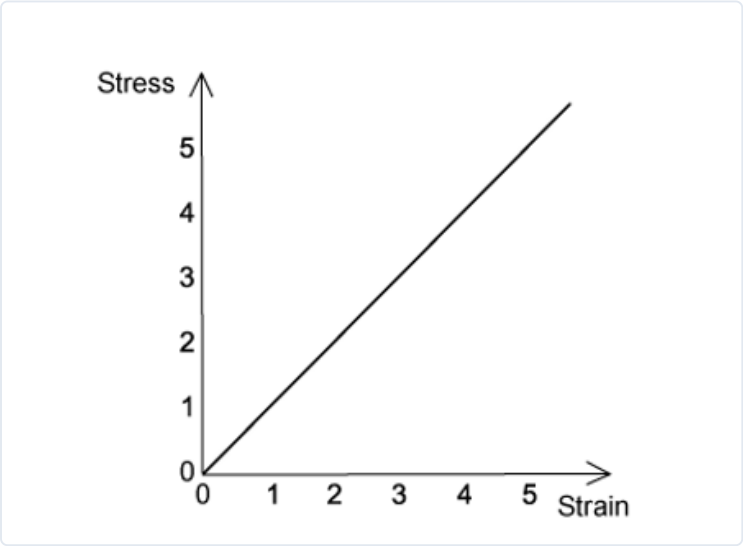
B 1.8%

C 4.7%

D 6.2%

Question 5 Easy [1 mark]

A glass rod was put under stress and strain until it broke. The graph of this is shown below.



Which of the following statements is correct about the glass rod?

A the glass shows plastic deformation

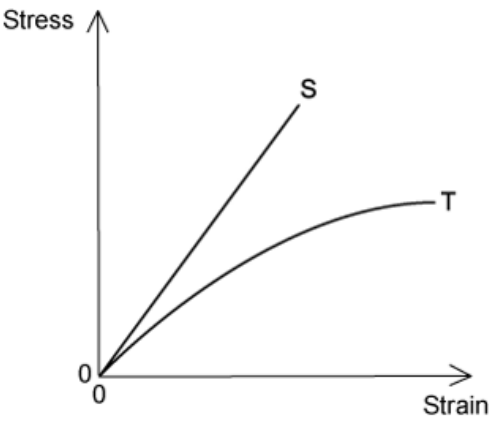
B when the cross-sectional area of the rod is doubled, the ultimate tensile stress of the rod is halved

C Hooke's law is obeyed for all values of stress up to the breaking point

D the glass is ductile

Question 6 Easy [1 mark]

Two wires S and T, with the same initial dimensions, were put under stress and strain, the graph shows the results up to their breaking points.



Which statement is **not** correct?

A material S has a larger ultimate tensile stress

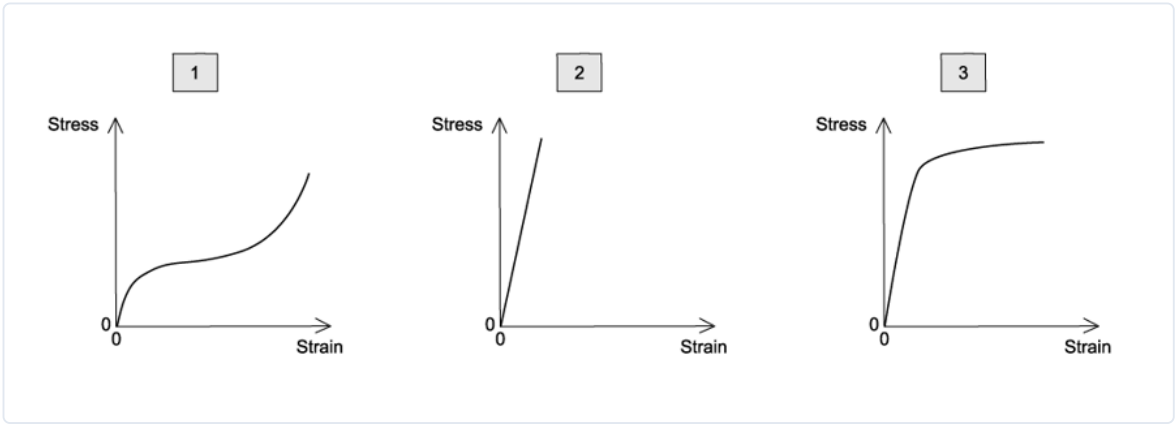
B material S extends elastically

C material S extends more than material T when loaded with the same force

D material S is brittle

Question 7 Easy [1 mark]

The graphs below show the stress-strain graphs of three materials. The graphs do not have the same scales.



The three materials are copper, rubber and glass.

Which materials are represented by the graphs?

	1	2	3
A	rubber	copper	glass
B	rubber	glass	copper
C	glass	copper	rubber
D	copper	glass	rubber

A

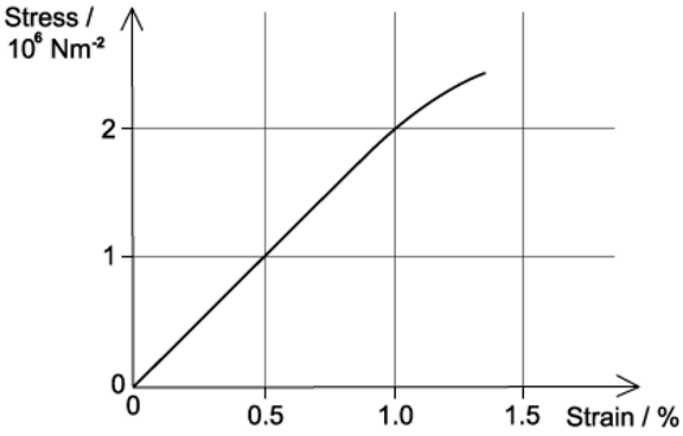
B

C

D

Question 8 Easy [1 mark]

The stress-strain graph for bone is shown below.



What is the Young modulus of bone?

A 2×10^6 N m^{-2}

B 2×10^8 N m^{-2}

C 1×10^6 N m^{-2}

D 1×10^8 N m^{-2}

Question 9 Easy [1 mark]

A student wanted to determine the spring constant of a spring. They did this by using different loads to extend the spring by varying amounts.

To find the spring constant, which quantities are required?

A the elastic limit, extensions and the length of the spring

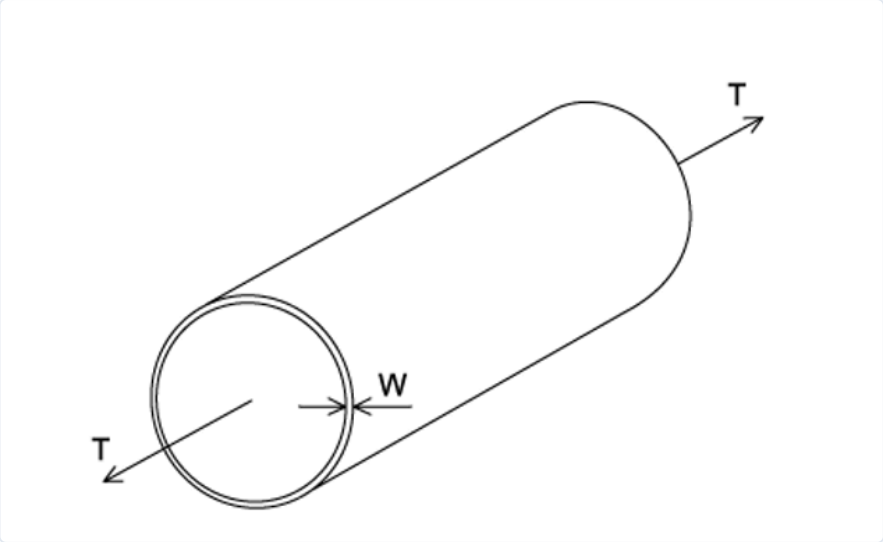
B the loads and the extensions of the spring

C the elastic limit and the loads

D the loads and the length of the spring

Question 10 Medium [1 mark]

A metal tube of diameter d with a thin wall thickness W is shown in the diagram. The thickness of the wall is small when compared to the diameter of the tube.



A force of magnitude T is applied parallel to the axis of the tube, putting the tube under tension. It is proposed that making the wall thicker would reduce the stress on the tube.

If the tube diameter and the tension remain the same, which wall thickness would half the stress?

A $\frac{W}{2}$

B $\sqrt{2}W$

C $2W$

D $4W$

Question 11 Medium [1 mark]

A steel metal wire has the following properties:

diameter = 5.0×10^{-4} m

Young modulus = 2.0×10^{11} Pa

tension = 20 N

The string snaps and contracts elastically.

By what percentage does a length *l* of a piece of the string contract?

A $1.3 \times 10^{-4}\,\%$

B $5.1 \times 10^{-4}\,\%$

C $1.3 \times 10^{-2}\,\%$

D $5.1 \times 10^{-2}\,\%$

Question 12 Medium [1 mark]

The Young modulus of a metal wire is dependent on which property?

A spring constant

B ductility

C ultimate tensile stress

D elastic limit

Question 13 Medium [1 mark]

The Young modulus *E* and the force per unit extension *k* describes the behaviour of a wire under tensile stress.

For a wire of length *L* and cross-sectional area *A*, what is the relationship between *E* and *k* ?

A $E = \frac{A}{kL}$

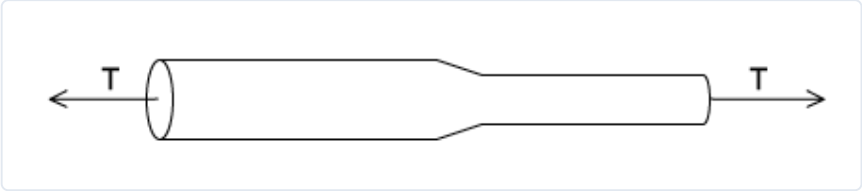
B $E = \frac{kA}{L}$

C $E = \frac{kL}{A}$

D $E = \frac{L}{kA}$

Question 14 Medium [1 mark]

A steel bar, shown in the diagram, has a circular cross-section that is under tension *T*.



The diameter of the thinner portion is half the wider portion.

What is the value of $\frac{stress\ \ on\ \ the\ \ wide\ \ portion}{stress\ \ on\ \ the\ \ narrow\ \ portion}$?

A 4.0

B 2.0

C 0.50

D 0.25

Question 15 Medium [1 mark]

A student wanted to find the young modulus of a wire. The equation for the young modulus *E* is:

$$E = \frac{4Fl}{\pi d^2 x}$$

The student extended the wire with a known force and made a series of measurements.

Which measurement has the largest effect on the uncertainty in the value of the calculated Young modulus?

	Measurement	Symbol	Value
A	length of wire before force applied	l	2.043 ± 0.002 m
B	force applied	F	19.62 ± 0.01 N
C	extension of wire with force applied	x	5.2 ± 0.2 mm
D	diameter of wire	d	0.54 ± 0.02 mm

A

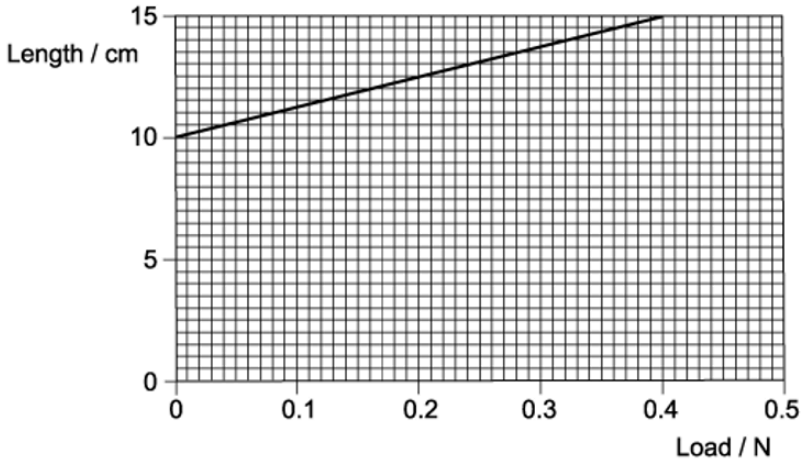
B

C

D

Question 16 Medium [1 mark]

A spring was stretched with increasing load. The graph of the results is shown below.



What is the spring constant?

A 0.080 N m^{-1}

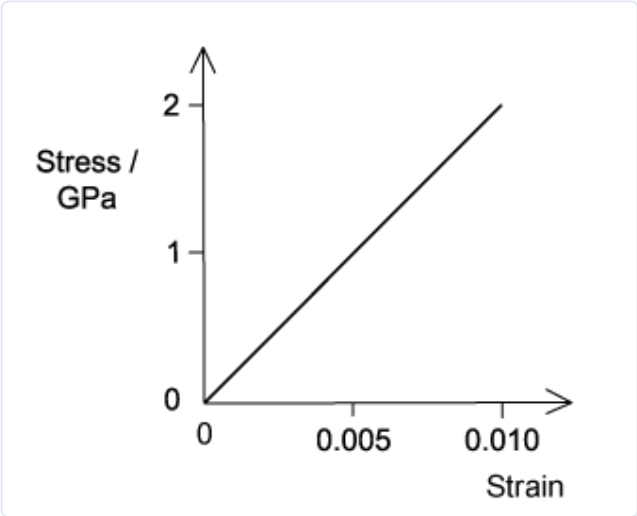
B 0.13 N m^{-1}

C 2.7 N m^{-1}

D 8.0 N m^{-1}

Question 17 Medium [1 mark]

The graph below shows the stress-strain for a metal.



When the strain on the rod is 0.010, what is the strain energy per unit volume of a rod made from this metal?

A 10 kJ m^{-3}

B 10 MJ m^{-3}

C 100 kJ m^{-3}

D 1.0 MJ m^{-3}

Question 18 Medium [1 mark]

A steel spring was stretched to a length of 55 cm when a 25 N weight was hung from it. The spring has a spring constant of 150 N m^{-1}.

What was the original length of the spring?

A 0.61 m

B 0.72 m

C 0.38 m

D 0.49 m

Question 19 Medium [1 mark]

A steel cable used in a construction crane has a Young's modulus of 2.1×10^{9} Pa. The cable has a cross-sectional area of 0.05 mm^2 and is subjected to a tensile force of 210 N. The cable obeys Hooke's law.

What is the stress in the steel cable?

A 4.2×10^{3} Pa

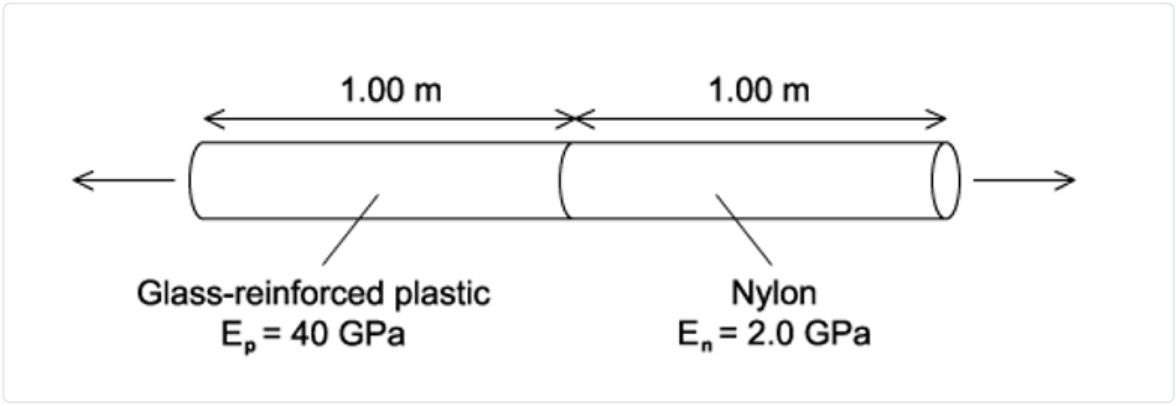
B 4.2×10^{6} Pa

C 4.2×10^{9} Pa

D 4.2×10^{12} Pa

Question 20 Hard [1 mark]

A glass-reinforced plastic rod and a nylon rod attached, as shown, was used to make a composite rod.



Each rod has a length of 1.00 m and the same cross-sectional area. The Young modulus E_n of the nylon is 2.0 GPa, and the Young modulus E_p of the plastic is 40 GPa

The composite rod will break when its total extension reaches 3.0 mm.

What is the greatest tensile stress that can be applied to the composite rod before it breaks?

A 5.7×10^{6} Pa

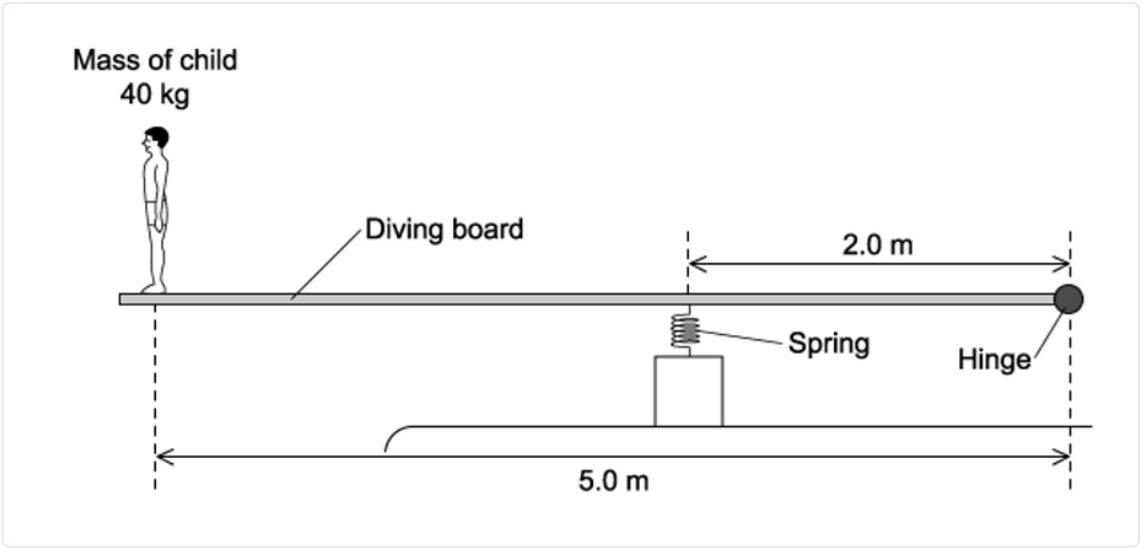
B 5.7×10^{9} Pa

C 7.1×10^{-14} Pa

D 7.1×10^{-2} Pa

Question 21 Hard [1 mark]

A child with a mass of 40 kg was standing on a diving board. The diving board has a length of 5.0 m and is hinged at one end then supported by a spring 2.0 m from the hinged end. The spring has a spring constant of 10 kN m^{-1}



What is the extra compression of the spring caused by the child standing on the end of the board?

A 16 cm

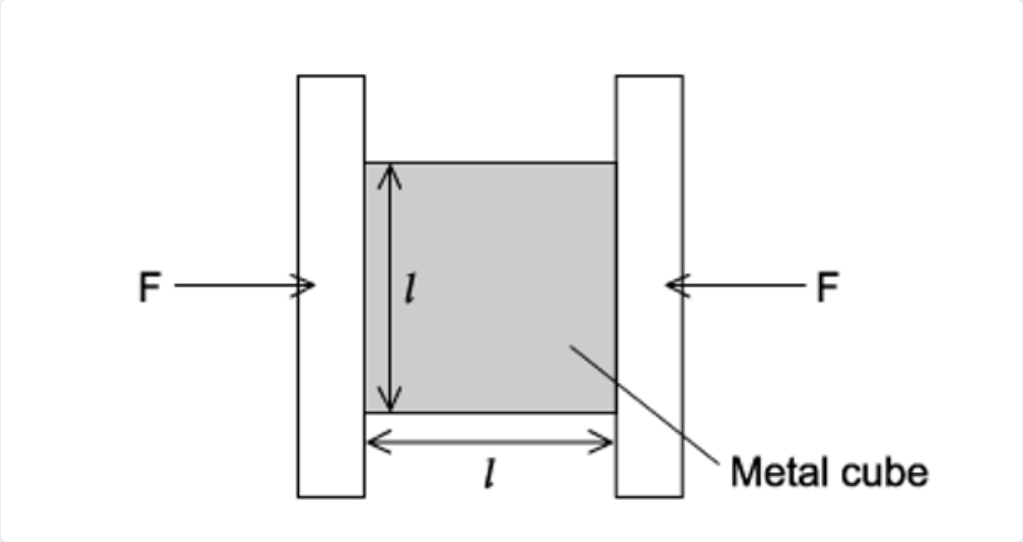
B 9.8 cm

C 1.6 cm

D 1.0 cm

Question 22 Hard [1 mark]

The diagram shows a metal cube is placed between two board and compressed elastically by two opposing forces *F*. The cube has a length of each side of l



How will Δl, the amount of compression, relate to l ?

A $\Delta l \ \alpha \ l$

B $\Delta l \ \alpha \ l^2$

C $\Delta l \ \alpha \ \frac{1}{l^2}$

D $\Delta l \ \alpha \ \frac{1}{l}$

Question 23 Hard [1 mark]

A metal wire that is supported vertically from a fixed point has a load of 84 N applied to the lower end. The wire has a cross-sectional area of 0.02 mm^2 and obeys Hooke's law. The length of the wire increases by 0.30 %.

What is the Young modulus of the metal wire?

A 1.4×10^5 Pa

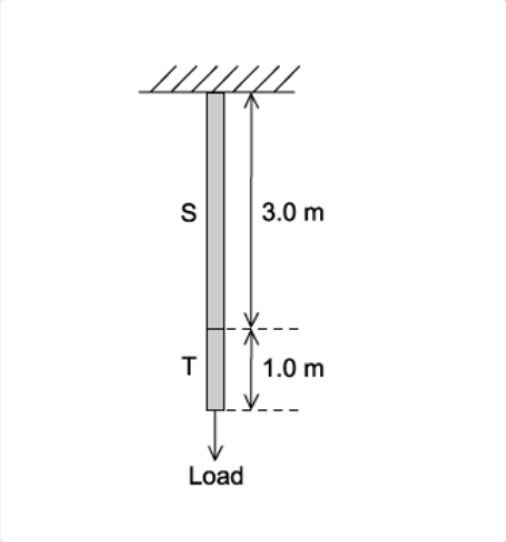
B 1.4×10^8 Pa

C 1.4×10^9 Pa

D 1.4×10^{12} Pa

Question 24 Hard [1 mark]

Two wires S and T is joined, S has a length of 3.0 m and T a length of 1.0 m, both wires have a uniform cross-sectional area.



When a load was hung from the wires S stretched by 1.5 mm and T by 1.0 mm.

The same load is then hung from a second wire of the same cross-section, consisting of 1.0 m of metal S and 3.0 m of metal T.

What is the total extension of this second wire?

A 3.5 mm

B 2.5 mm

C 5.0 mm

D 4.8 mm

Question 25 Hard [1 mark]

Two steel wires X and Y that both obey Hooke's law were used in an experiment. Wire X has a length l and a cross-sectional area A. Wire Y has a length of $2l$ and cross-sectional area of $\frac{A}{2}$.

What is the ratio $\frac{tension\ \ in\ \ X}{tension\ \ in\ \ Y}$ when both wires are stretched to the same extension?

A $\frac{1}{4}$

B $\frac{4}{1}$

C $\frac{2}{1}$

D $\frac{1}{2}$